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Electrical and Electronics Series

Circuits & Networks

Analysis and Synthesis

A Sudhakar
Shyammohan S Palli

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Chapter 1

Circuit Elements and Kirchhoff's Laws

1.1 VOLTAGE

According to the structure of an atom, we know that there are two types of charges: positive and negative. A force of attraction exists between these positive and negative charges. A certain amount of energy (work) is required to overcome the force and move the charges through a specific distance. All opposite charges possess a certain amount of potential energy because of the separation between them. The difference in potential energy of the charges is called the *potential difference*.

Potential difference in electrical terminology is known as voltage, and is denoted either by V or v. It is expressed in terms of energy (W) per unit charge (Q); i.e.

$$V = \frac{W}{Q} \quad \text{or} \quad v = \frac{dw}{dq}$$

dw is the small change in energy, and
dq is the small change in charge.

where energy (W) is expressed in joules (J), charge (Q) in coulombs (C), and voltage (V) in volts (V). One volt is the potential difference between two points when one joule of energy is used to pass one coulomb of charge from one point to the other.

Example 1.1 If 70 J of energy is available for every 30 C of charge, what is the voltage?

Solution $$V = \frac{W}{Q} = \frac{70}{30} = 2.33 \text{ V}$$

1.2 CURRENT

There are free electrons available in all semiconductive and conductive materials. These free electrons move at random in all directions within the structure in the absence of external pressure or voltage. If a certain amount of voltage is applied across the material, all the free electrons move in one direction depending on the polarity of the applied voltage, as shown in Fig. 1.1.

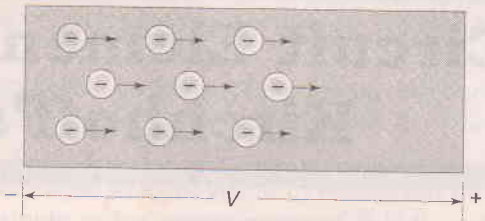

Fig. 1.1

This movement of electrons from one end of the material to the other end constitutes an electric current, denoted by either I or i. The conventional direction of current flow is opposite to the flow of – ve charges, i.e. the electrons.

Current is defined as the rate of flow of electrons in a conductive or semiconductive material. It is measured by the number of electrons that flow past a point in unit time. Expressed mathematically,

$$I = \frac{Q}{t}$$

where I is the current, Q is the charge of electrons, and t is the time, or

$$i = \frac{dq}{dt}$$

where dq is the small change in charge, and dt is the small change in time.

In practice, the unit *ampere* is used to measure current, denoted by A. One ampere is equal to one coulomb per second. One coulomb is the charge carried by 6.25×10^{18} electrons. For example, an ordinary 80 W domestic ceiling fan on 230 V supply takes a current of approximately 0.35 A. This means that electricity is passing through the fan at the rate of 0.35 coulomb every second, i.e. 2.187×10^{18} electrons are passing through the fan in every second; or simply, the current is 0.35 A.

Example 1.2 Five coulombs of charge flow past a given point in a wire in 2 s. How many amperes of current is flowing?

Solution $I = \frac{Q}{t} = \frac{5}{2} = 2.5$ A

1.3 POWER AND ENERGY

Energy is the capacity for doing work, i.e. energy is nothing but stored work. Energy may exist in many forms such as mechanical, chemical, electrical and so

on. Power is the rate of c
certain amount of energy i

Power (P) =

p =

where dw is the change in

We can also write p =

=

Energy is measured in j
By definition, one watt
energy is consumed in one
second is always equal to t
watt are usually expressed
milliwatts (mW) and micr
and megawatts (MW) are
large amounts of power to

Example 1.3 What is th
2.5 s?

Solution P =

1.4 THE CIRCUIT

Simply an electric circuit
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cannot flow, the circuit is
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More specifically, inte
two or more simple circuit
voltage sources, resistors,
capacitors) is called an ele

on. Power is the rate of change of energy, and is denoted by either P or p. If certain amount of energy is used over a certain length of time, then

$$\text{Power } (P) = \frac{\text{energy}}{\text{time}} = \frac{W}{t}, \text{ or}$$

$$p = \frac{dw}{dt}$$

where dw is the change in energy and dt is the change in time.

We can also write $$p = \frac{dw}{dt} = \frac{dw}{dq} \times \frac{dq}{dt}$$

$$= v \times i = vi \quad \text{W}$$

Energy is measured in joules (J), time in seconds (s), and power in watts (W).

By definition, one watt is the amount of power generated when one joule of energy is consumed in one second. Thus, the number of joules consumed in one second is always equal to the number of watts. Amounts of power less than one watt are usually expressed in fraction of watts in the field of electronics; viz. milliwatts (mW) and microwatts (μW). In the electrical field, kilowatts (kW) and megawatts (MW) are common units. Radio and television stations also use large amounts of power to transmit signals.

Example 1.3 What is the power in watts if energy equal to 50 J is used in 2.5 s?

Solution $$P = \frac{\text{energy}}{\text{time}} = \frac{50}{2.5} = 20 \text{ W}$$

1.4 THE CIRCUIT

Simply an electric circuit consists of three parts: (1) energy source, such as battery or generator, (2) the load or sink, such as lamp or motor, and (3) connecting wires as shown in Fig. 1.2. This arrangement represents a simple circuit. A battery is connected to a lamp with two wires. The purpose of the circuit is to transfer energy from source (battery) to the load (lamp). And this is accomplished by the passage of electrons through wires around the circuit.

The current flows through the filament of the lamp, causing it to emit visible light. The current flows through the battery by chemical action. A closed circuit is defined as a circuit in which the current has a complete path to flow. When the current path is broken so that current cannot flow, the circuit is called an open circuit.

More specifically, interconnection of two or more simple circuit elements (viz. voltage sources, resistors, inductors and capacitors) is called an electric network.

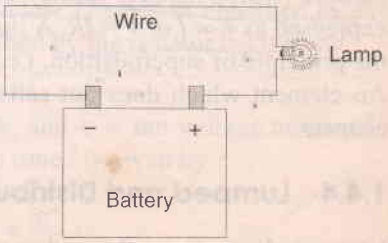

Fig. 1.2

If a network contains at least one closed path, it is called an electric circuit. By definition, a simple circuit element is the mathematical model of two terminal electrical devices, and it can be completely characterised by its voltage and current. Evidently then, a physical circuit must provide means for the transfer of energy.

Broadly, network elements may be classified into four groups, viz.

1. Active or passive
2. Unilateral or bilateral
3. Linear or nonlinear
4. Lumped or distributed

1.4.1 Active and Passive

Energy sources (voltage or current sources) are active elements, capable of delivering power to some external device. Passive elements are those which are capable only of receiving power. Some passive elements like inductors and capacitors are capable of storing a finite amount of energy, and return it later to an external element. More specifically, an active element is capable of delivering an average power greater than zero to some external device over an infinite time interval. For example, ideal sources are active elements. A passive element is defined as one that cannot supply average power that is greater than zero over an infinite time interval. Resistors, capacitors and inductors fall into this category.

1.4.2 Bilateral and Unilateral

In the bilateral element, the voltage-current relation is the same for current flowing in either direction. In contrast, a unilateral element has different relations between voltage and current for the two possible directions of current. Examples of bilateral elements are elements made of high conductivity materials in general. Vacuum diodes, silicon diodes, and metal rectifiers are examples of unilateral elements.

1.4.3 Linear and Nonlinear Elements

An element is said to be linear, if its voltage-current characteristic is at all times a straight line through the origin. For example, the current passing through a resistor is proportional to the voltage applied through it, and the relation is expressed as $V \propto I$ or $V = IR$. A linear element or network is one which satisfies the principle of superposition, i.e. the principle of homogeneity and additivity. An element which does not satisfy the above principle is called a nonlinear element.

1.4.4 Lumped and Distributed

Lumped elements are those elements which are very small in size and in which simultaneous actions takes place for any given cause at the same instant of time.

Typical lumped elements a
Generally the elements are
compared to the wave leng
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For example, a transmissio
and capacitance along its le

1.5 RESISTANCE PAR

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resistance, denoted by R.
resistor is shown in Fig. 1.

The unit of resistance is
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According to Ohm's lav
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$I =$

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We can write the above equ

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When current flows thr
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$P =$

where i is the current in th
resistor in volts. Energy los

$W =$

Typical lumped elements are capacitors, resistors, inductors and transformers. Generally the elements are considered as lumped when their size is very small compared to the wave length of the applied signal. Distributed elements, on the other hand, are those which are not electrically separable for analytical purposes. For example, a transmission line which has distributed resistance, inductance and capacitance along its length may extend for hundreds of miles.

1.5 RESISTANCE PARAMETER

When a current flows in a material, the free electrons move through the material and collide with other atoms. These collisions cause the electrons to lose some of their energy. This loss of energy per unit charge is the drop in potential across the material. The amount of energy lost by the electrons is related to the physical property of the material. These collisions restrict the movement of electrons. The property of a material to restrict the flow of electrons is called resistance, denoted by R. The symbol for the resistor is shown in Fig. 1.3.

Fig. 1.3

The unit of resistance is ohm (Ω). Ohm is defined as the resistance offered by the material when a current of one ampere flows between two terminals with one volt applied across it.

According to Ohm's law, the current is directly proportional to the voltage and inversely proportional to the total resistance of the circuit, i.e.

$$I = \frac{V}{R}$$

or

$$i = \frac{v}{R}$$

We can write the above equation in terms of charge as follows.

$$V = R\frac{dq}{dt}, \quad \text{or} \quad i = \frac{v}{R} = Gv$$

where G is the conductance of a conductor. The units of resistance and conductance are ohm (Ω) and mho (℧) respectively.

When current flows through any resistive material, heat is generated by the collision of electrons with other atomic particles. The power absorbed by the resistor is converted to heat. The power absorbed by the resistor is given by

$$P = vi = (iR)i = i^2 R$$

where i is the current in the resistor in amps, and v is the voltage across the resistor in volts. Energy lost in a resistance in time t is given by

$$W = \int_0^t p\,dt = pt = i^2 Rt = \frac{v^2}{R}t$$

where v is the volts
R is in ohms
t is in seconds and
W is in joules

Example 1.4 A 10 Ω resistor is connected across a 12 V battery. How much current flows through the resistor?

Solution $V = IR$

$$I = \frac{V}{R} = \frac{12}{10} = 1.2 \text{ A}$$

1.6 INDUCTANCE PARAMETER

A wire of certain length, when twisted into a coil becomes a basic inductor. If current is made to pass through an inductor, an electromagnetic field is formed. A change in the magnitude of the current changes the electromagnetic field. Increase in current expands the fields, and decrease in current reduces it. Therefore, a change in current produces change in the electromagnetic field, which induces a voltage across the coil according to Faraday's law of electromagnetic induction.

The unit of inductance is *henry*, denoted by H. By definition, the inductance is one henry when current through the coil, changing at the rate of one ampere per second, induces one volt across the coil. The symbol for inductance is shown in Fig. 1.4.

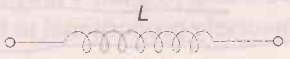

Fig. 1.4

The current-voltage relation is given by

$$v = L\frac{di}{dt}$$

where v is the voltage across inductor in volts, and i is the current through inductor in amps. We can rewrite the above equations as

$$di = \frac{1}{L}\,v\,dt$$

Integrating both sides, we get

$$\int_0^t di = \frac{1}{L}\int_0^t v\,dt$$

$$i(t) - i(0) = \frac{1}{L}\int_0^t v\,dt$$

$$i(t) = \frac{1}{L}\int_0^t v\,dt + i(0)$$

From the above equation
upon the integral of the volta
coil, $i(0)$.

The power absorbed by in

$P =$

The energy stored by the i

$W =$

From the above discussion,

1. The induced voltage a
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Example 1.5 The current
voltage across the inductor

Solution $v =$

$=$

$W =$

$=$

1.7 CAPACITANCE P

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From the above equation we note that the current in an inductor is dependent upon the integral of the voltage across its terminals and the initial current in the coil, $i(0)$.

The power absorbed by inductor is

$$P = vi = Li\frac{di}{dt} \text{ watts}$$

The energy stored by the inductor is

$$W = \int_0^t p\,dt$$

$$= \int_0^t Li\frac{di}{dt}dt = \frac{Li^2}{2}$$

From the above discussion, we can conclude the following.

1. The induced voltage across an inductor is zero if the current through it is constant. That means an inductor acts as short circuit to dc.
2. A small change in current within zero time through an inductor gives an infinite voltage across the inductor, which is physically impossible. In a fixed inductor the current cannot change abruptly.
3. The inductor can store finite amount of energy, even if the voltage across the inductor is zero, and
4. A pure inductor never dissipates energy, only stores it. That is why it is also called a non-dissipative passive element. However, physical inductors dissipate power due to internal resistance.

Example 1.5 The current in a 2 H inductor varies at a rate of 2 A/s. Find the voltage across the inductor and the energy stored in the magnetic field after 2 s.

Solution

$$v = L\frac{di}{dt}$$

$$= 2 \times 4 = 8 \text{ V}$$

$$W = \frac{1}{2}Li^2$$

$$= \frac{1}{2} \times 2 \times (4)^2 = 16 \text{ J}$$

1.7 CAPACITANCE PARAMETER

Any two conducting surfaces separated by an insulating medium exhibit the property of a capacitor. The conducting surfaces are called *electrodes*, and the insulating medium is called *dielectric*. A capacitor stores energy in the form of an electric field that is established by the opposite charges on the two electrodes. The electric field is represented by lines of force between the positive and negative charges, and is concentrated within the dielectric. The amount of charge

per unit voltage that is capacitor can store is its capacitance, denoted by C. The unit of capacitance is *Farad* denoted by F. By definition, one Farad is the amount of capacitance when one coulomb of charge is stored with one volt across the plates. The symbol for capacitance is shown in Fig. 1.5.

Fig. 1.5

A capacitor is said to have greater capacitance if it can store more charge per unit voltage and the capacitance is given by

$$C = \frac{Q}{V}, \quad \text{or} \quad C = \frac{q}{v}$$

(lower case letters stress instantaneous values)

We can write the above equation in terms of current as

$$i = C\frac{dv}{dt} \qquad \left(\because \; i = \frac{dq}{dt}\right)$$

where v is the voltage across capacitor, i is the current through it

$$dv = \frac{1}{C}\, idt$$

Integrating both sides, we have

$$\int_0^t dv = \frac{1}{C}\int_0^t idt$$

$$v(t) - v(0) = \frac{1}{C}\int_0^t idt$$

$$v(t) = \frac{1}{C}\int_0^t idt + v(0)$$

where $v(0)$ indicates the initial voltage across the capacitor.

From the above equation, the voltage in a capacitor is dependent upon the integral of the current through it, and the initial voltage across it.

The power absorbed by the capacitor is given by

$$p = vi = vC\frac{dv}{dt}$$

The energy stored by the capacitor is

$$W = \int_0^t pdt = \int_0^t vC\frac{dv}{dt}dt$$

$$W = \frac{1}{2}Cv^2$$

From the above discussio

1. The current in a ca
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2. A small change in
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Example 1.6 A capacit
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Solution W

1.8 ENERGY SOURC

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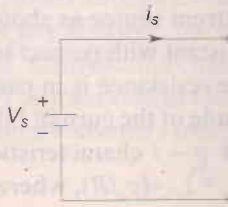

(a)

If we observe the $v - i$
Fig. 1.6(c) at any time,
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From the above discussion we can conclude the following

1. The current in a capacitor is zero if the voltage across it is constant; that means, the capacitor acts as an open circuit to dc.
2. A small change in voltage across a capacitance within zero time gives an infinite current through the capacitor, which is physically impossible. In a fixed capacitance the voltage cannot change abruptly.
3. The capacitor can store a finite amount of energy, even if the current through it is zero, and
4. A pure capacitor never dissipates energy, but only stores it; that is why it is called *non-dissipative passive element*. However, physical capacitors dissipate power due to internal resistance.

Example 1.6 A capacitor having a capacitance 2 μF is charged to a voltage of 1000 V. Calculate the stored energy in joules.

Solution $$W = \frac{1}{2} Cv^2 = \frac{1}{2} \times 2 \times 10^{-6} \times (1000)^2 = 1 \text{ J}.$$

1.8 ENERGY SOURCES

According to their terminal voltage-current characteristics, electrical energy sources are categorised into ideal voltage sources and ideal current sources. Further they can be divided into independent and dependent sources.

An ideal voltage source is a two-terminal element in which the voltage v_s is completely independent of the current i_s through its terminals. The representation of ideal constant voltage source is shown in Fig. 1.6(a).

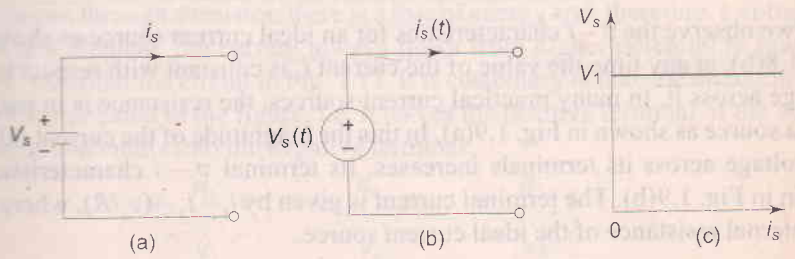

Fig. 1.6

If we observe the $v - i$ characteristics for an ideal voltage source as shown in Fig. 1.6(c) at any time, the value of the terminal voltage v_s is constant with respect to the value of current i_s. Whenever $v_s = 0$, the voltage source is the same as that of a short circuit. Voltage sources need not have constant magnitude; in many cases the specified voltage may be time-dependent like a sinusoidal waveform. This may be represented as shown in Fig. 1.6(b). In many practical voltage sources, the internal resistance is represented in series with the source as shown in Fig. 1.7(a). In this, the voltage across the terminals falls as the current through it increases, as shown in Fig. 1.7 (b).

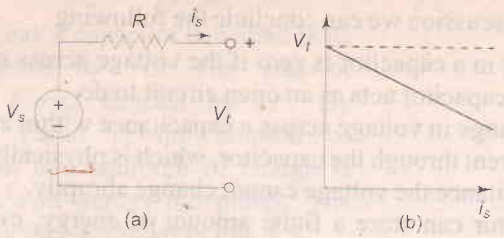

Fig. 1.7

The terminal voltage v_t depends on the source current as shown in Fig. 1.7(b), where $v_t = v_s - i_s R$.

An ideal constant current source is a two-terminal element in which the current i_s completely independent of the voltage v_s across its terminals. Like voltage sources we can have current sources of constant magnitude i_s or sources whose current varies with time $i_s(t)$. The representation of an ideal current source is shown in Fig. 1.8(a).

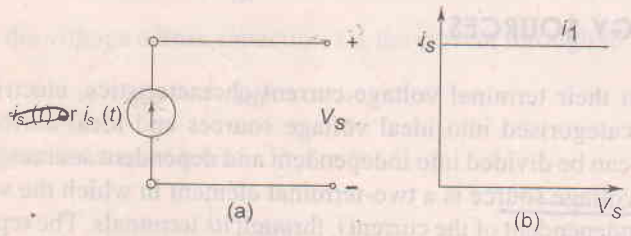

Fig. 1.8

If we observe the $v - i$ characteristics for an ideal current source as shown in Fig. 1.8(b), at any time the value of the current i_s is constant with respect to the voltage across it. In many practical current sources, the resistance is in parallel with a source as shown in Fig. 1.9(a). In this the magnitude of the current falls as the voltage across its terminals increases. Its terminal $v - i$ characteristics is shown in Fig. 1.9(b). The terminal current is given by $i_t = i_s - (v_s/R)$, where R is the internal resistance of the ideal current source.

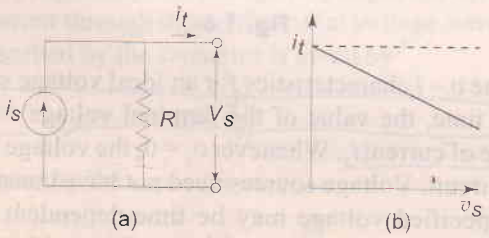

Fig. 1.9

The two types of ideal sources we have discussed are independent sources for which voltage and current are independent and are not affected by other parts of

the circuit. In the case of dependent sources, the source voltage or current is not fixed, but is dependent on the voltage or current existing at some other location in the circuit.

Dependent or controlled sources are of the following types

(i) voltage controlled voltage source (VCVS)
(ii) current controlled voltage source (CCVS)
(iii) voltage controlled current source (VCCS)
(iv) current controlled current source (CCCS)

These are represented in a circuit diagram by the symbol shown in Fig. 1.10. These types of sources mainly occur in the analysis of equivalent circuits of transistors.

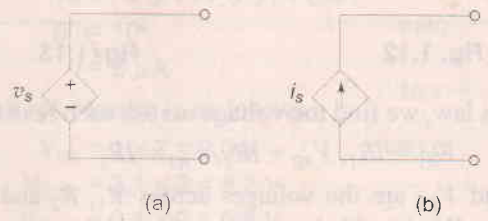

Fig. 1.10

1.9 KIRCHHOFF'S VOLTAGE LAW

Kirchhoff's voltage law states that the algebraic sum of all branch voltages around any closed path in a circuit is always zero at all instants of time. When the current passes through a resistor, there is a loss of energy and, therefore, a voltage drop. In any element, the current always flows from higher potential to lower potential. Consider the circuit in Fig. 1.11. It is customary to take the direction of current I as indicated in the figure, i.e. it leaves the positive terminal of the voltage source and enters into the negative terminal.

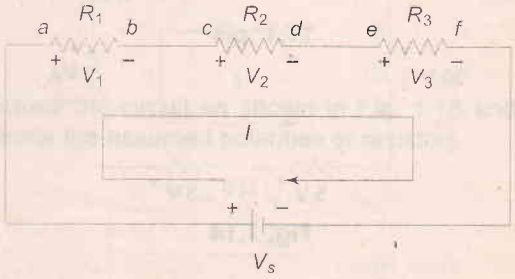

Fig. 1.11

As the current passes through the circuit, the sum of the voltage drop around the loop is equal to the total voltage in that loop. Here the polarities are attributed to the resistors to indicate that the voltages at points a, c and e are more than the voltages at b, d and f, respectively, as the current passes from a to f.

$$\therefore \quad V_s = V_1 + V_2 + V_3$$

Consider the problem of finding out the current supplied by the source V in the circuit shown in Fig. 1.12.

Our first step is to assume the reference current direction and to indicate the polarities for different elements. (See Fig. 1.13).

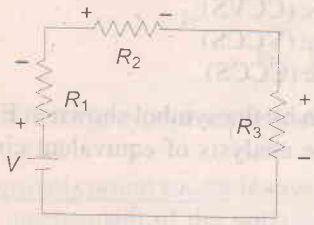

Fig. 1.12

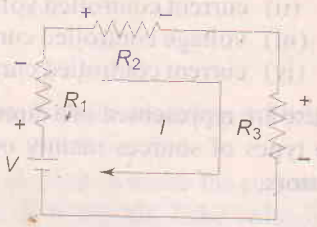

Fig. 1.13

By using Ohm's law, we find the voltage across each resistor as follows.

$$V_{R1} = IR_1,\ V_{R2} = IR_2,\ V_{R3} = IR_3$$

where V_{R1}, V_{R2} and V_{R3} are the voltages across R_1, R_2 and R_3, respectively. Finally, by applying Kirchhoff's law, we can form the equation

$$V = V_{R1} + V_{R2} + V_{R3}$$

$$V = IR_1 + IR_2 + IR_3$$

From the above equation the current delivered by the source is given by

$$I = \frac{V}{R_1 + R_2 + R_3}$$

Example 1.7 For the circuit shown in Fig. 1.14, determine the unknown voltage drop V_1.

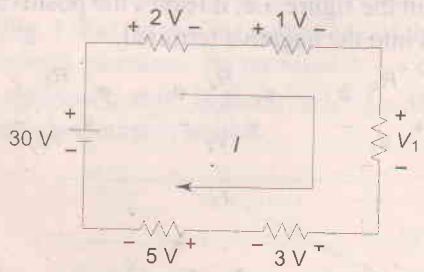

Fig. 1.14

Solution According to Kirchhoff's voltage law, the sum of the potential drops is equal to the sum of the potential rises;

Therefore, $30 = 2 + 1 + V_1 + 3 + 5$

or $V_1 = 30 - 11 = 19$ V

Example 1.8 What is the current in the circuit shown in Fig. 1.15? Determine the voltage across each resistor.

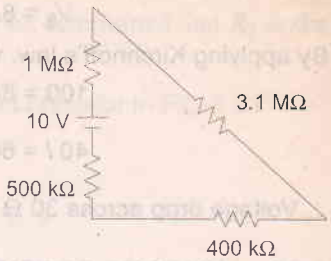

Fig. 1.15

Solution We assume current I in the clockwise direction and indicate polarities (Fig. 1.16). By using Ohm's law, we find the voltage drops across each resistor.

$$V_{1M} = I, \qquad V_{3.1M} = 3.1I$$
$$V_{500K} = 0.5I, \qquad V_{400K} = 0.4I$$

Now, by applying Kirchhoff's voltage law, we form the equation.

$$10 = I + 3.1\,I + 0.5\,I + 0.4\,I$$

or $$5\,I = 10$$

or $$I = 2\ \mu A$$

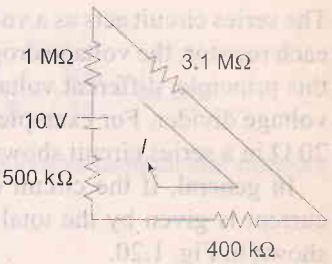

Fig. 1.16

∴ Voltage across each resistor is as follows

$$V_{1M} = 1 \times 2 = 2.0\ V$$
$$V_{3.1M} = 3.1 \times 2 = 6.2\ V$$
$$V_{400K} = 0.4 \times 2 = 0.8\ V$$
$$V_{500K} = 0.5 \times 2 = 1.0\ V$$

Example 1.9 In the circuit given in Fig. 1.17, find (a) the current I, and (b) the voltage across 30 Ω.

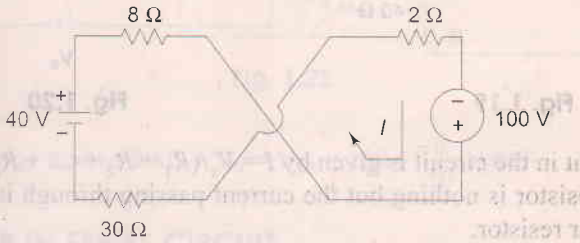

Fig. 1.17

Solution We redraw the circuit as shown in Fig. 1.18 and assume current direction and indicate the assumed polarities of resistors

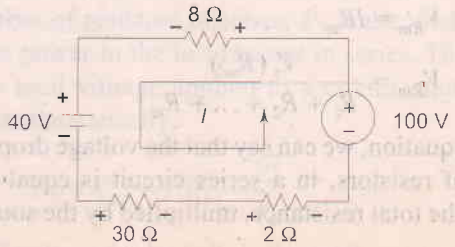

Fig. 1.18

By using Ohm's law, we determine the voltage across each resistor as

$$V_8 = 8I,\ V_{30} = 30I,\ V_2 = 2I$$

By applying Kirchhoff's law, we get

$$100 = 8I + 40 + 30I + 2I$$

$$40\,I = 60 \text{ or } I = \frac{60}{40} = 1.5 \text{ A}$$

$\therefore$ Voltage drop across 30 Ω = V_{30} = 30 × 1.5 = 45 V

1.10 VOLTAGE DIVISION

The series circuit acts as a voltage divider. Since the same current flows through each resistor, the voltage drops are proportional to the values of resistors. Using this principle, different voltages can be obtained from a single source, called a voltage divider. For example, the voltage across a 40 Ω resistor is twice that of 20 Ω in a series circuit shown in Fig. 1.19.

In general, if the circuit consists of a number of series resistors, the total current is given by the total voltage divided by equivalent resistance. This is shown in Fig. 1.20.

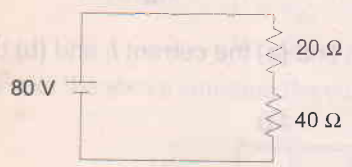

Fig. 1.19

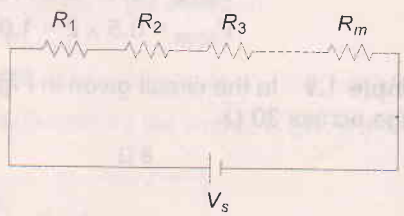

Fig. 1.20

The current in the circuit is given by $I = V_s/(R_1 + R_2 + \ldots + R_m)$. The voltage across any resistor is nothing but the current passing through it, multiplied by that particular resistor.

Therefore,

$$V_{R1} = IR_1$$
$$V_{R2} = IR_2$$
$$V_{R3} = IR_3$$
$$\vdots$$
$$V_{Rm} = IR_m$$

or

$$V_{Rm} = \frac{V_s(R_m)}{R_1 + R_2 + \ldots + R_m}$$

From the above equation, we can say that the voltage drop across any resistor, or a combination of resistors, in a series circuit is equal to the ratio of that resistance value to the total resistance, multiplied by the source voltage, i.e.

$$V_m = \frac{R_m}{R_T} V_s$$

where V_m is the voltage across mth resistor,
R_m is the resistance across which the voltage is to be determined and R_T is the total series resistance.

Example 1.10 What is the voltage across the 10 Ω resistor in Fig. 1.21.

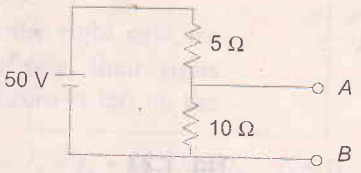

Fig. 1.21

Solution Voltage across 10 Ω = $V_{10} = 50 \times \frac{10}{10+5} = \frac{500}{15} = 33.3$ V

Example 1.11 Find the voltage between A and B in a voltage divider network shown in Fig. 1.22.

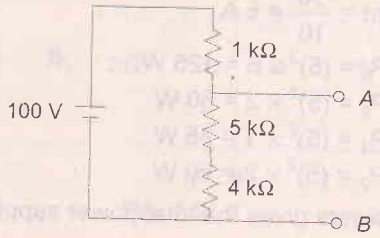

Fig. 1.22

Solution Voltage across 9 kΩ = $V_9 = V_{AB} = 100 \times \frac{9}{10} = 90$ V

1.11 POWER IN SERIES CIRCUIT

The total power supplied by the source in any series resistive circuit is equal to the sum of the powers in each resistor in series, i.e.

$$P_S = P_1 + P_2 + P_3 + \ldots + P_m$$

where m is the number of resistors in series, P_S is the total power supplied by source and P_m is the power in the last resistor in series. The total power in the series circuit is the total voltage applied to a circuit, multiplied by the total current. Expressed mathematically,

$$P_S = V_s I = I^2 R_T = \frac{V_s^2}{R_T}$$

where V_s is the total voltage applied, R_T is the total resistance, and I is the total current.

Example 1.12 Determine the total amount of power in the series circuit in Fig. 1.23.

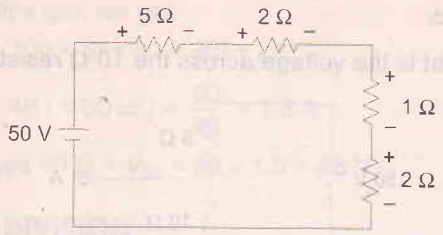

Fig. 1.23

Solution Total resistance = 5 + 2 + 1 + 2 = 10 Ω

We know $$P_S = \frac{V_s^2}{R_T} = \frac{(50)^2}{10} = 250 \text{ W}$$

Check We find the power absorbed by each resistor

$$\text{Current} = \frac{50}{10} = 5 \text{ A}$$

$$P_5 = (5)^2 \times 5 = 125 \text{ W}$$
$$P_2 = (5)^2 \times 2 = 50 \text{ W}$$
$$P_1 = (5)^2 \times 1 = 25 \text{ W}$$
$$P_2 = (5)^2 \times 2 = 50 \text{ W}$$

The sum of these powers gives the total power supplied by the source P_S = 250 W.

1.12 KIRCHHOFF'S CURRENT LAW

Kirchhoff's current law states that the sum of the currents entering into any node is equal to the sum of the currents leaving that node. The node may be an interconnection of two or more branches. In any parallel circuit, the node is a junction point of two or more branches. The total current entering into a node is equal to the current leaving that node. For example, consider the circuit shown in Fig. 1.24, which contains two nodes A and B. The total current I_T entering node A is divided into I_1, I_2 and I_3. These currents flow out of node A. According to Kirchhoff's current law, the current into node A is equal to the total current out of node A: that is, $I_T = I_1 + I_2 + I_3$. If we consider node B, all three currents I_1, I_2, I_3 are entering B, and the total current I_T is leaving node B, Kirchhoff's current law formula at this node is therefore the same as at node A.

$$I_1 + I_2 + I_3 = I_T$$

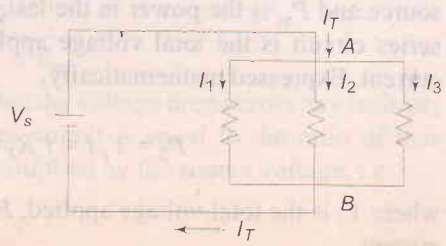

Fig. 1.24

In general, sum of the currents entering any point or node or junction equal to sum of the currents leaving from that point or node or junction as shown in Fig. 1.25.

$$I_1 + I_2 + I_4 + I_7 = I_3 + I_5 + I_6$$

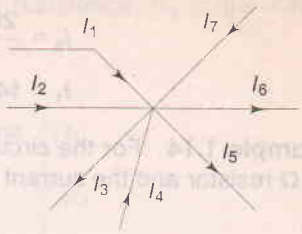

Fig. 1.25

If all of the terms on the right side are brought over to the left side, their signs change to negative and a zero is left on the right side, i.e.

$$I_1 + I_2 + I_4 + I_7 - I_3 - I_5 - I_6 = 0$$

This means that the algebraic sum of all the currents meeting at a junction is equal to zero.

Example 1.13 Determine the current in all resistors in the circuit shown in Fig. 1.26.

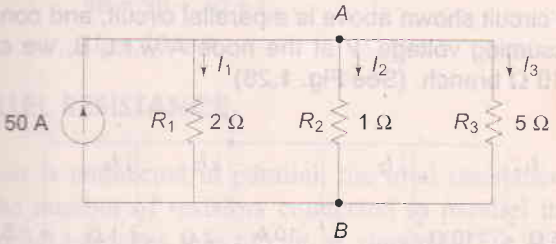

Fig. 1.26

Solution The above circuit contains a single node 'A' with reference node 'B'. Our first step is to assume the voltage V at node A. In a parallel circuit the same voltage is applied across each element. According to Ohm's law, the currents passing through each element are $I_1 = V/2$, $I_2 = V/1$, $I_3 = V/5$.

By applying Kirchhoff's current law, we have

$$I = I_1 + I_2 + I_3$$

$$I = \frac{V}{2} + \frac{V}{1} + \frac{V}{5}$$

$$50 = V\left[\frac{1}{2} + \frac{1}{1} + \frac{1}{5}\right] = V\,[0.5 + 1 + 0.2]$$

$$V = \frac{50}{1.7} = \frac{500}{17} = 29.41 \text{ V}$$

Once we know the voltage V at node A, we can find the current in any element by using Ohm's law.

The current in the 2 Ω resistor is $I_1 = 29.41/2 = 14.705$ A.

Similarly $$I_2 = \frac{V}{R_2} = \frac{V}{1} = 29.41 \text{ A}$$

$$I_3 = \frac{29.41}{5} = 5.882 \text{ A}$$

$\therefore$ $I_1 = 14.7$ A, $I_2 = 29.4$ A, and $I_3 = 5.88$ A

Example 1.14 For the circuit shown in Fig. 1.27, find the voltage across the 10 Ω resistor and the current passing through it.

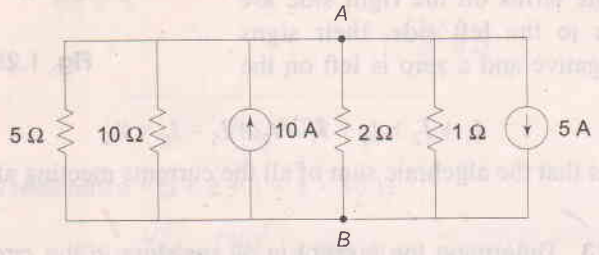

Fig. 1.27

Solution The circuit shown above is a parallel circuit, and consists of a single node A. By assuming voltage V at the node A w.r.t. B, we can find out the current in the 10 Ω branch. (See Fig. 1.28)

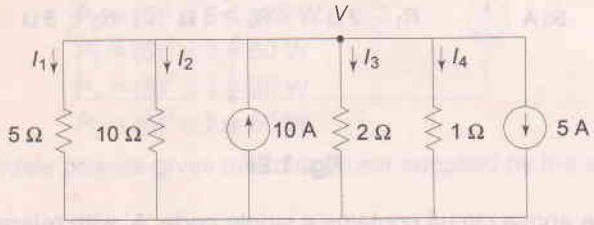

Fig. 1.28

According to Kirchhoff's current law,

$$I_1 + I_2 + I_3 + I_4 + 5 = 10$$

By using Ohm's law we have

$$I_1 = \frac{V}{5};\ I_2 = \frac{V}{10},\ I_3 = \frac{V}{2},\ I_4 = \frac{V}{1}$$

$$\frac{V}{5} + \frac{V}{10} + \frac{V}{2} + V + 5 = 10$$

$$V\left[\frac{1}{5} + \frac{1}{10} + \frac{1}{2} + 1\right] = 5$$

$$V\,[0.2 + 0.1 + 0.5 + 1] = 5$$

$$V = \frac{5}{1.8} = 2.78 \text{ V}$$

$\therefore$ The voltage across the 10 Ω resistor is 2.78 V and the current passing through it is

$$I_2 = \frac{V}{10} = \frac{2.78}{10} = 0.278 \text{ A}$$

Example 1.15 Determine the current through resistance R_3 in the circuit shown in Fig. 1.29.

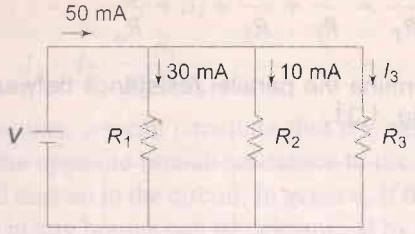

Fig. 1.29

Solution According to Kirchhoff's current law,

$$I_T = I_1 + I_2 + I_3$$

where I_T is the total current and I_1, I_2 and I_3 are the currents in resistances R_1, R_2 and R_3 respectively.

$$50 = 30 + 10 + I_3$$

or

$$I_3 = 10 \text{ mA}$$

1.13 PARALLEL RESISTANCE

When the circuit is connected in parallel, the total resistance of the circuit decreases as the number of resistors connected in parallel increases. If we consider m parallel branches in a circuit as shown in Fig. 1.30, the current equation is

$$I_T = I_1 + I_2 + \ldots + I_m$$

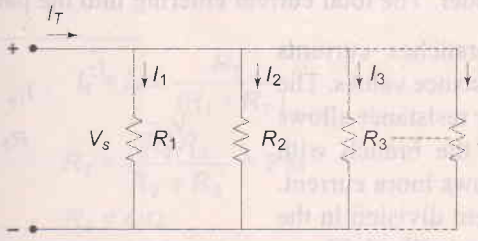

Fig. 1.30

The same voltage is applied across each resistor. By applying Ohm's law, the current in each branch is given by

$$I_1 = \frac{V_s}{R_1},\ I_2 = \frac{V_s}{R_2},\ \ldots\ I_m = \frac{V_s}{R_m}$$

According to Kirchhoff's current law,

$$I_T = I_1 + I_2 + I_3 + \ldots + I_m$$

$$\frac{V_s}{R_T} = \frac{V_s}{R_1} + \frac{V_s}{R_2} + \frac{V_s}{R_3} + \ldots + \frac{V_s}{R_m}$$

From the above equation, we have

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_m}$$

Example 1.16 Determine the parallel resistance between points A and B of the circuit shown in Fig. 1.31.

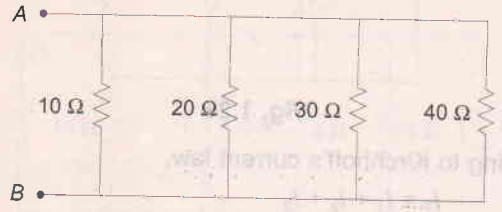

Fig. 1.31

Solution

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$

$$\frac{1}{R_T} = \frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{40}$$

$$= 0.1 + 0.05 + 0.033 + 0.025 = 0.208$$

or $R_T = 4.8\ \Omega$

1.14 CURRENT DIVISION

In a parallel circuit, the current divides in all branches. Thus, a parallel circuit acts as a current divider. The total current entering into the parallel branches is divided into the branches currents according to the resistance values. The branch having higher resistance allows lesser current, and the branch with lower resistance allows more current. Let us find the current division in the parallel circuit shown in Fig. 1.32.

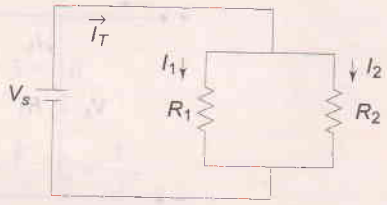

Fig. 1.32

The voltage applied across each resistor is V_s. The current passing through each resistor is given by

$$I_1 = \frac{V_s}{R_1},\ I_2 = \frac{V_s}{R_2}$$

If R_T is the total resistance, which is given by $R_1R_2/(R_1 + R_2)$,

Total current $$I_T = \frac{V_s}{R_T} = \frac{V_s}{R_1 R_2}(R_1 + R_2)$$

or $$I_T = \frac{I_1 R_1}{R_1 R_2}(R_1 + R_2) \text{ since } V_s = I_1 R_1$$

$$I_1 = I_T \cdot \frac{R_2}{R_1 + R_2}$$

Similarly,

$$I_2 = I_T \cdot \frac{R_1}{R_1 + R_2}$$

From the above equations, we can conclude that the current in any branch is equal to the ratio of the opposite branch resistance to the total resistance value, multiplied by the total current in the circuit. In general, if the circuit consists of m branches, the current in any branch can be determined by

$$I_i = \frac{R_T}{R_i + R_T} I_T$$

where I_i represents the current in the ith branch
R_i is the resistance in the ith branch
R_T is the total parallel resistance to the ith branch and
I_T is the total current entering the circuit.

Example 1.17 Determine the current through each resistor in the circuit shown in Fig. 1.33.

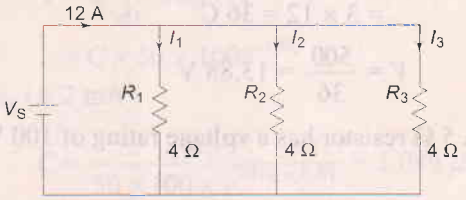

Fig. 1.33

Solution

$$I_1 = I_T \times \frac{R_T}{(R_1 + R_T)}$$

where

$$R_T = \frac{R_2 R_3}{R_2 + R_3} = 2\ \Omega$$

$$R_1 = 4\ \Omega$$

$$I_T = 12\ \text{A}$$

$$I_1 = 12 \times \frac{2}{2+4} = 4\ \text{A}$$

Similarly,

$$I_2 = 12 \times \frac{2}{2+4} = 4\ \text{A}$$

and

$$I_3 = 12 \times \frac{2}{2+4} = 4\ \text{A}$$

Since all parallel branches have equal values of resistance, they share current equally.

1.15 POWER IN PARALLEL CIRCUIT

The total power supplied by the source in any parallel resistive circuit is equal to the sum of the powers in each resistor in parallel, i.e.

$$P_S = P_1 + P_2 + P_3 + \ldots + P_m$$

where m is the number of resistors in parallel, P_S is the total power and P_m is the power in the last resistor.

Additional Solved Problems

Problem 1.1 A resistor with a current of 3 A through it converts 500 J of electrical energy to heat energy in 12 s. What is the voltage across the resistor?

Solution

$$V = \frac{W}{Q}$$

$$Q = I \times t$$

$$= 3 \times 12 = 36 \text{ C}$$

$$V = \frac{500}{36} = 13.88 \text{ V}$$

Problem 1.2 A 5 Ω resistor has a voltage rating of 100 V. What is its power rating?

Solution

$$P = VI$$

$$I = V/R$$

$$P = \frac{V^2}{R} = \frac{(100)^2}{5} = 2000 \text{ W} = 2 \text{ kW}$$

Problem 1.3 Find the inductance of a coil through which flows a current of 0.2 A with an energy of 0.15 J.

Solution

$$W = \frac{1}{2} LI^2$$

$$L = \frac{2 \times W}{I^2} = \frac{2 \times 0.15}{(0.2)^2} = 7.5 \text{ H}$$

Problem 1.4 Find the inductance of a coil in which a current increases linearly from 0 to 0.2 A in 0.3 s, producing a voltage of 15 V.

Solution

$$v = L\frac{di}{dt}$$

Current in 1 second $= \dfrac{0.2}{0.3} = 0.66$ A

The current changes at a rate of 0.66 A/s,

$$\therefore \quad L = \frac{v}{\left(\frac{di}{dt}\right)}$$

$$L = \frac{15\,\text{V}}{0.66\,\text{A/s}} = 22.73\ \text{H}$$

Problem 1.5 When a dc voltage is applied to a capacitor, the voltage across its terminals is found to build up in accordance with $v_C = 50(1 - e^{-100t})$. After a lapse of 0.01 s, the current flow is equal to 2 mA.

(a) Find the value of capacitance in microfarads

(b) How much energy is stored in the electric field at this time?

Solution

(a)
$$i = C\frac{dv_C}{dt}$$

where $v_C = 50(1 - e^{-100t})$

$$i = C\frac{d}{dt}\,50(1 - e^{-100t})$$

$$= C \times 50 \times 100e^{-100t}$$

At $t = 0.01$ s, $i = 2$ mA

$$C = \frac{2 \times 10^{-3}}{50 \times 100 \times e^{-100 \times 0.01}} = 1.089\ \mu\text{F}$$

(b)
$$W = \frac{1}{2}\,Cv_C^2$$

At $t = 0.01$ s, $v_C = 50\,(1 - e^{-100 \times 0.01}) = 31.6$ V

$$W = \frac{1}{2} \times 1.089 \times 10^{-6} \times (31.6)^2$$

$$= 0.000543\ \text{J}$$

Problem 1.6 Determine the total current in the circuit shown in Fig. 1.34.

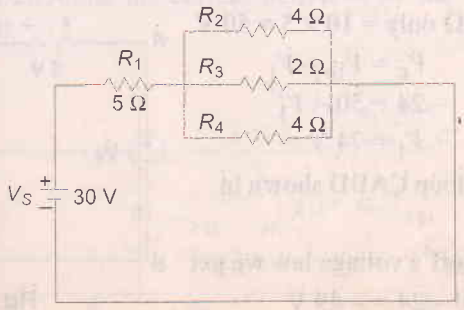

Fig. 1.34

Solution Resistances R_2, R_3 and R_4 are in parallel

$\therefore$ Equivalent resistance $R_5 = R_2 \parallel R_3 \parallel R_4$

$$= \frac{1}{1/R_2 + 1/R_3 + 1/R_4}$$

$\therefore$ $R_5 = 1\ \Omega$

R_1 and R_5 are in series,

$\therefore$ Equivalent resistance $R_T = R_1 + R_5 = 5 + 1 = 6\ \Omega$

And the total current $I_T = \dfrac{V_s}{R_T} = \dfrac{30}{6} = 5$ A

Problem 1.7 Find the current in the 10 Ω resistance, V_1, and source voltage V_s in the circuit shown in Fig. 1.35.

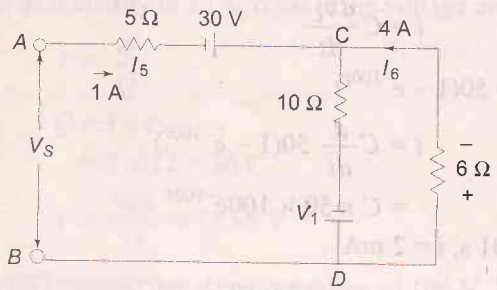

Fig. 1.35

Solution Assume voltage at node $C = V$

By applying Kirchhoff's current law, we get the current in the 10 Ω resistance

$$I_{10} = I_5 + I_6$$
$$= 4 + 1 = 5 \text{ A}$$

The voltage across the 6 Ω resistor is $V_6 = 24$ V

$\therefore$ Voltage at node C is $V_C = -24$ V.

The voltage across branch CD is the same as the voltage at node C.

Voltage across 10 Ω only = 10 × 5 = 50 V

So $V_C = V_{10} - V_1$

$-24 = 50 - V_1$

$\therefore$ $V_1 = 74$ V

Now, consider the loop CABD shown in Fig. 1.36

If we apply Kirchhoff's voltage law we get

$V_s = 5 - 30 - 24 = -49$ V

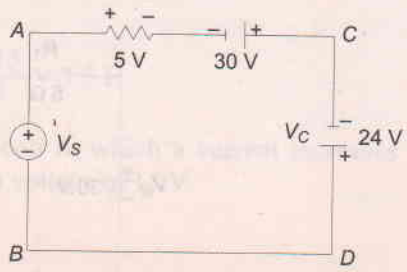

Fig. 1.36